Differentiation

McMaster University

Mathematics and Statistics

$$\int_M d\omega = \int_{\partial M} \omega$$

Mathematics 3A03
Real Analysis I

Instructor: David Earn

Lecture 2
Differentiation
Wednesday 8 January 2025

Survey to do right now

- Please go to
  https://www.childsmath.ca/childsa/forms/main_login.php
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Announcements

Announcements

- Results of Survey 1

Announcements

- Results of Survey 1
- Results of Survey 2

Background / reminder

Definition (Cauchy sequence)

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A sequence $\{s_n\}$

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A sequence $\{s_n\}$ is said to be a ***Cauchy sequence*** iff for all $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that if $m \geq N$ and $n \geq N$ then $|s_n - s_m| < \varepsilon$.

Poll: another background check

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  https://www.childsmath.ca/childsa/forms/main_login.php
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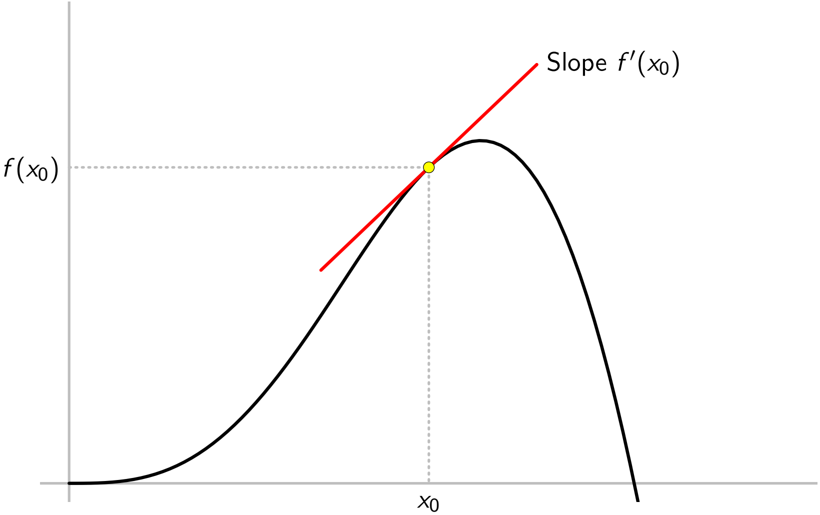
- Fill in poll **Background: Cauchy sequences**
- $\boxed{\texttt{Submit}}$.

The Derivative

The Derivative



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- Two things are equal, but in one $x \neq 2$ and in the other $x = 2$.
- Good illustration of why it is important to define the meaning of limits rigorously.

Poll

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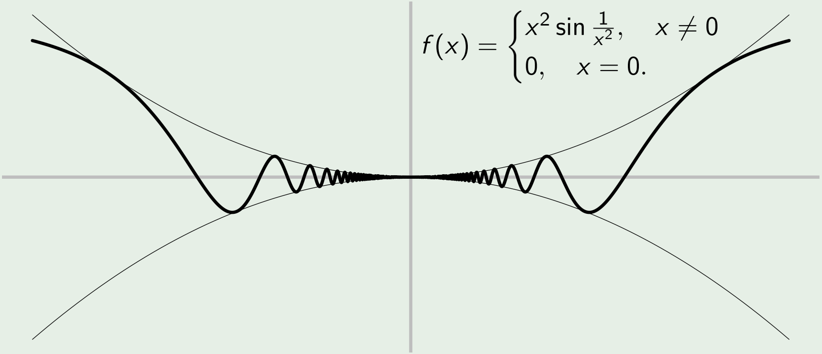

$$f(x) = \begin{cases} x^2 \sin \frac{1}{x^2}, & x \neq 0 \\ 0, & x = 0. \end{cases}$$

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$\therefore$ f is differentiable at 0 and $f'(0) = 0$. $\square$

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Let f be defined on an interval I and let $x_0 \in I$. The ***right-hand derivative*** of f at x_0, denoted by $f'_+(x_0)$, is the limit

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Note: If x_0 is not an endpoint of the interval I then f is differentiable at x_0 iff $f'_+(x_0) = f'_-(x_0) \neq \pm\infty$.

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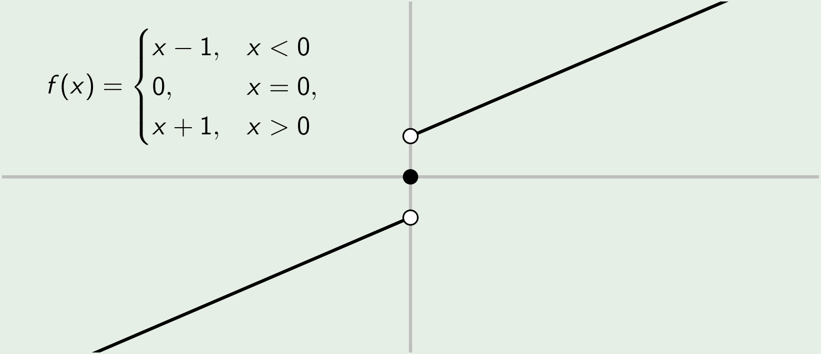
$$f(x) = \begin{cases} x - 1, & x < 0 \\ 0, & x = 0, \\ x + 1, & x > 0 \end{cases}$$



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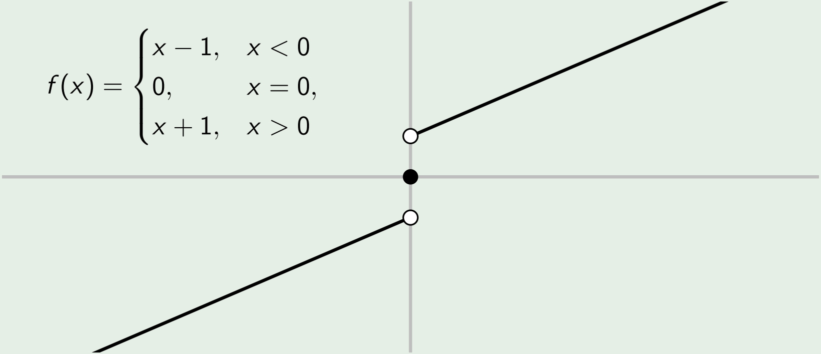

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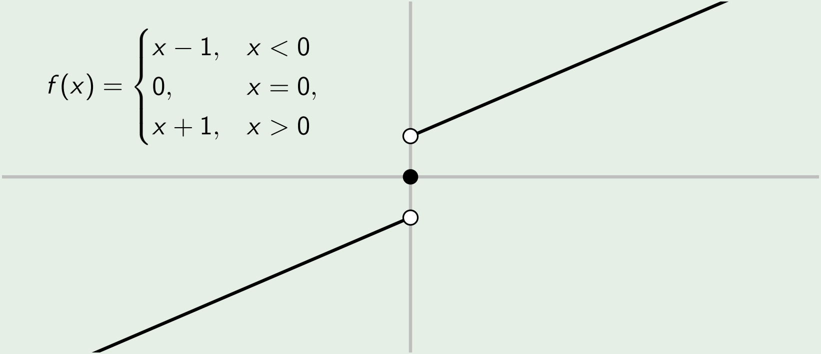

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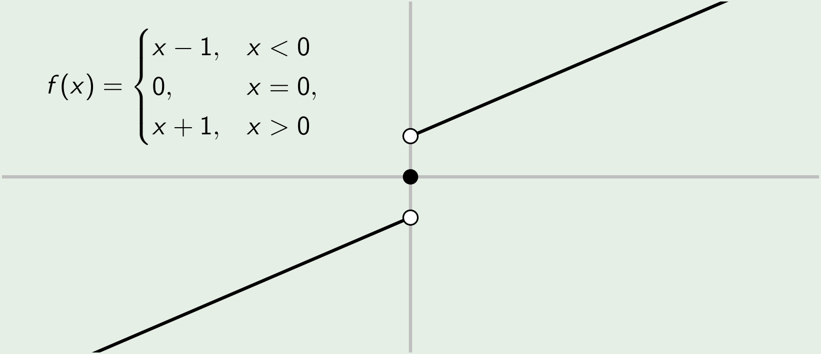

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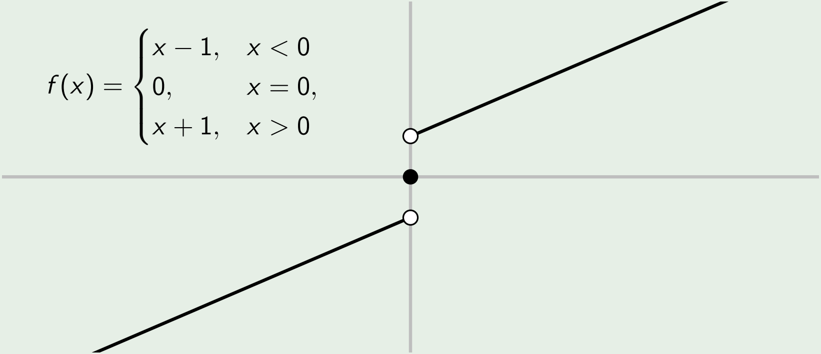

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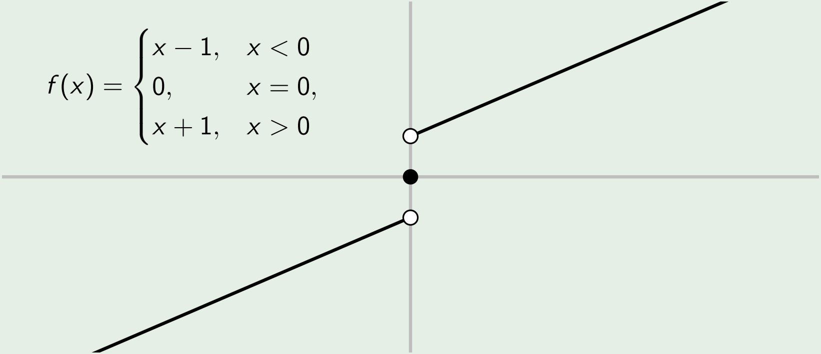

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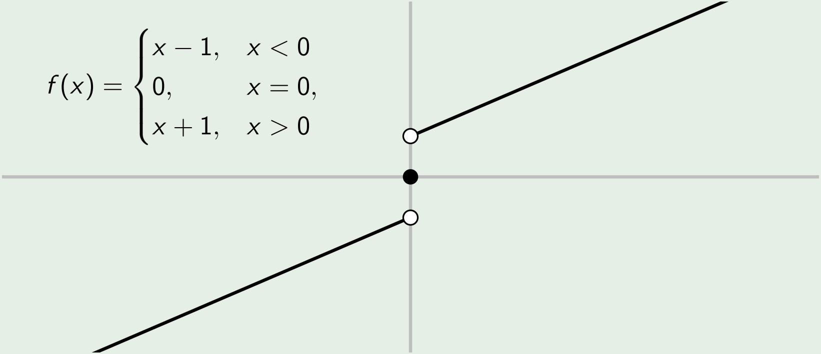

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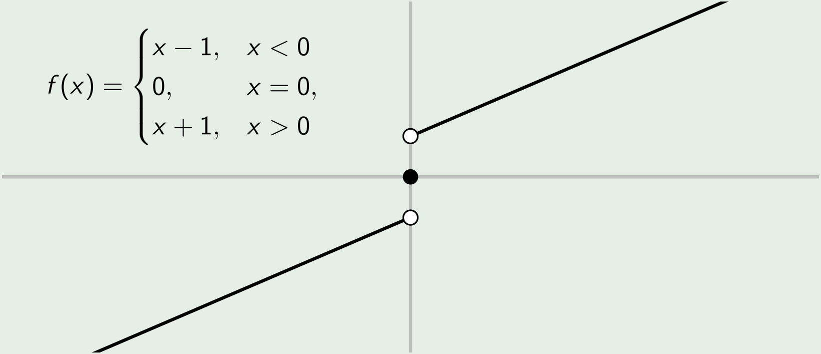

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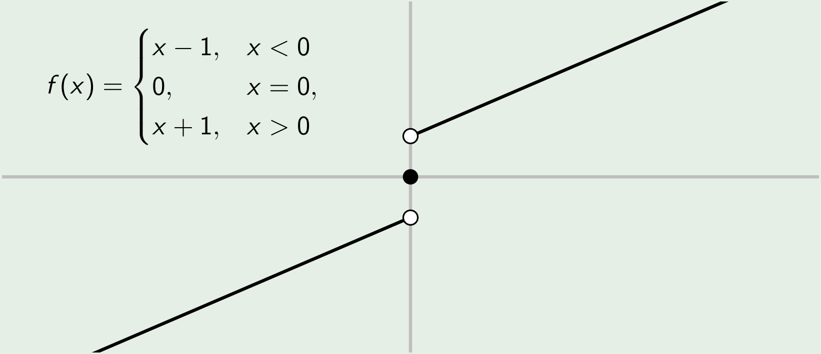

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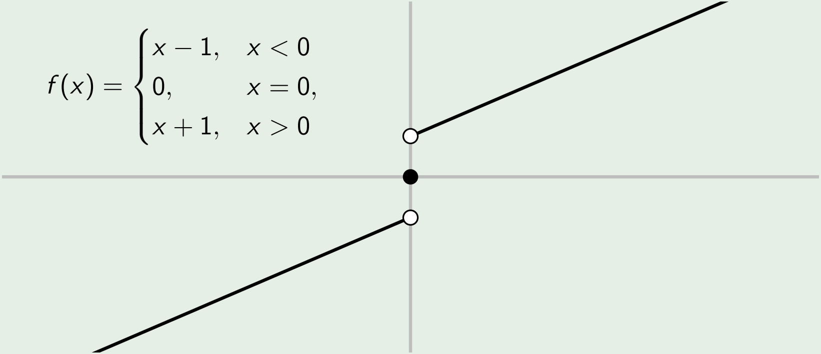

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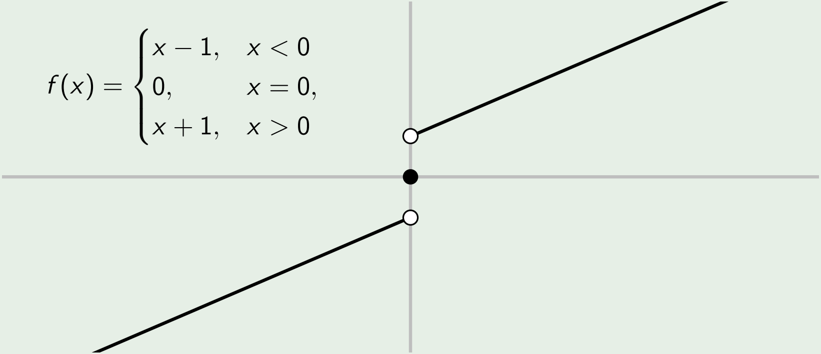

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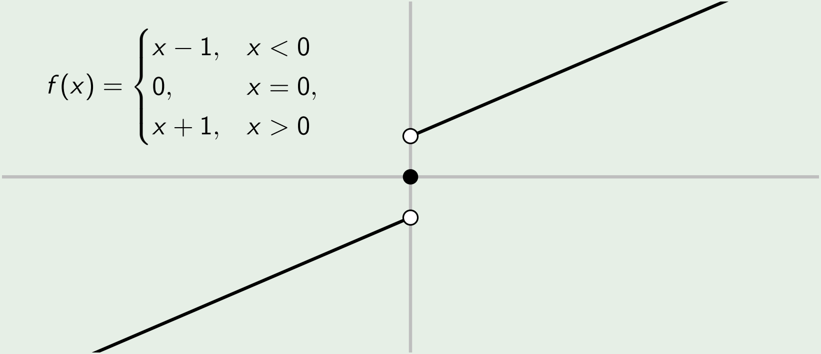

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$$D^n f(x) = \frac{d^n f}{dx^n} = f^{(n)}(x)$$

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5. *if* $t_n \neq 0$ *for all* n *and* $\lim_{n\to\infty} t_n \neq 0$ *then*
$$\lim_{n\to\infty}\left(\frac{s_n}{t_n}\right) = \frac{\lim_{n\to\infty} s_n}{\lim_{n\to\infty} t_n} .$$

(TBB §2.7, and problem 2.7.4)

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5. *if* $g(x) \neq 0$ *for* $x \in (x_0 - \delta, x_0 + \delta)$ *for some* $\delta > 0$, *and* $\lim_{x\to x_0} g(x) \neq 0$ *then* $\lim\limits_{x\to x_0} \left(\dfrac{f(x)}{g(x)}\right) = \dfrac{\lim_{x\to x_0} f(x)}{\lim_{x\to x_0} g(x)}$.

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&= \lim_{x\to x_0} \left(\frac{f(x)-f(x_0)}{x-x_0}\right) \times \lim_{x\to x_0} (x-x_0) \\
&= f'(x_0) \times 0 =
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\lim_{x\to x_0} \big(f(x) - f(x_0)\big) &= \lim_{x\to x_0} \left(\frac{f(x)-f(x_0)}{x-x_0} \times (x-x_0)\right) \\
&= \lim_{x\to x_0} \left(\frac{f(x)-f(x_0)}{x-x_0}\right) \times \lim_{x\to x_0} (x-x_0) \\
&= f'(x_0) \times 0 = 0\,,
\end{aligned}$$

The Derivative

Theorem (Differentiable $\implies$ continuous)

If f is defined in a neighbourhood I of x_0 and f is differentiable at x_0 then f is continuous at x_0.

Proof.

Must show $\lim_{x \to x_0} f(x) = f(x_0)$, *i.e.*, $\lim_{x \to x_0} (f(x) - f(x_0)) = 0$.

$$\begin{aligned}
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where we have used the theorem on the algebra of limits. $\square$

McMaster University

Mathematics and Statistics

$$\int_M d\omega = \int_{\partial M} \omega$$

Mathematics 3A03
Real Analysis I

Instructor: David Earn

Lecture 3
Differentiation II
Friday 10 January 2025

Announcements

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  - Select "MATH 3A03 - WINTER 2025"
- Course evaluation scheme has changed (next two slides).
  - The course web site and online syllabus have not yet been updated to reflect these changes, but that will happen soon (hopefully over the weekend).

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Last time. . .

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- Definition of the derivative.

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More on the derivative

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4. $\left(\frac{f}{g}\right)'(x_0) = \left(\frac{gf'-fg'}{g^2}\right)(x_0) \qquad (g(x_0) \neq 0)$.

(TBB Theorem 7.7, p. 408)

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(TBB §7.3.2, p. 411)

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- Can we use $(\spadesuit)$ to prove the chain rule?

Poll

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- Click on Take Class Poll
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- Submit.

REMINDER: limits of functions

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Note: The deleted neighbourhood ($I \setminus \{x_0\}$) can be replaced by any set on which f is defined and x_0 is an accumulation point.

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Note: TBB's proof leaves out the proof that $f'(x_0) = 0$ in case 2 above. □

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Theorem (Derivative at local extrema)

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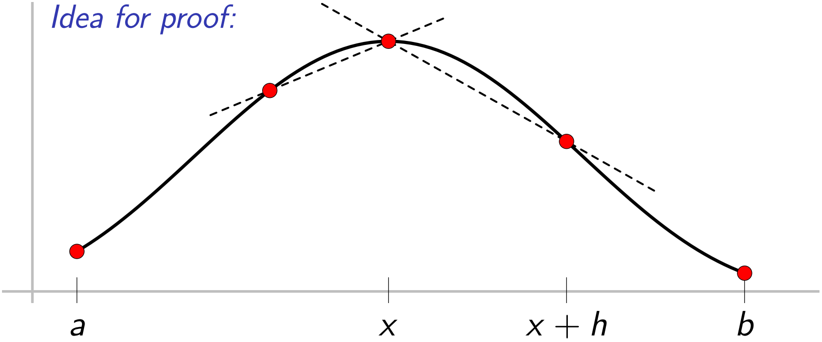
Idea for proof:



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$$f'_+(x) \leq 0 \leq f'_-(x) .$$

But since f is differentiable at x, the left and right derivatives must be equal, hence $f'(x) = 0$. □

McMaster University

Mathematics and Statistics

$$\int_M d\omega = \int_{\partial M} \omega$$

Mathematics 3A03
Real Analysis I

Instructor: David Earn

Lecture 4
Differentiation III
Monday 13 January 2025

Announcements

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- In-Class polls:

Announcements

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- The course web site has been updated to reflect the changes in how the course will be evaluated.

Assignment 1

Assignment 1

- Assignment 1 has been posted on the course web site.

Last time. . .

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- Discussed algebra of derivatives and chain rule.

Last time. . .

- Discussed algebra of derivatives and chain rule.
- Proved the chain rule.

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- Discussed algebra of derivatives and chain rule.
- Proved the chain rule.
- Proved that that derivative is zero at extrema.

The Mean Value Theorem

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Theorem (Rolle's theorem)

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Proof.

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f continuous on $[a,b]$ $\implies$

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f continuous on $[a, b] \implies f$ has a max and min value

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$$f'(x) = \frac{f(b) - f(a)}{b - a}\,.$$

The Mean Value Theorem

The Mean Value Theorem

Idea for proof:

The Mean Value Theorem

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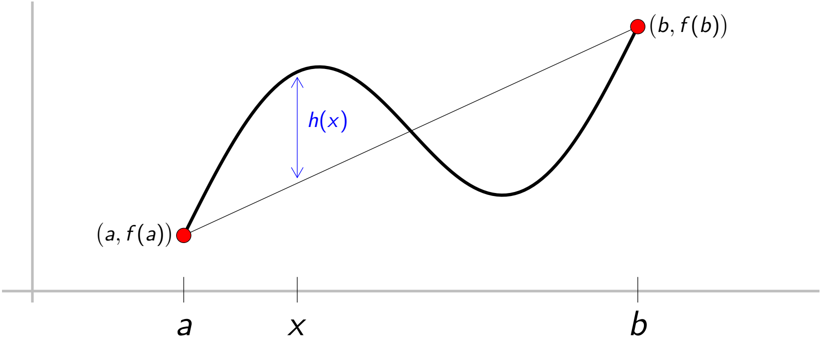

The Mean Value Theorem

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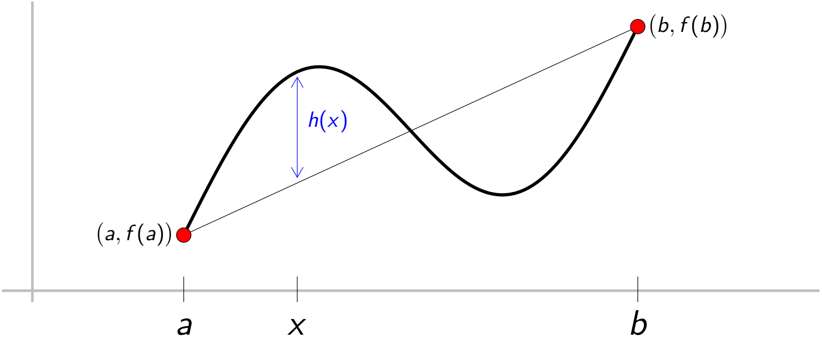

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The Mean Value Theorem

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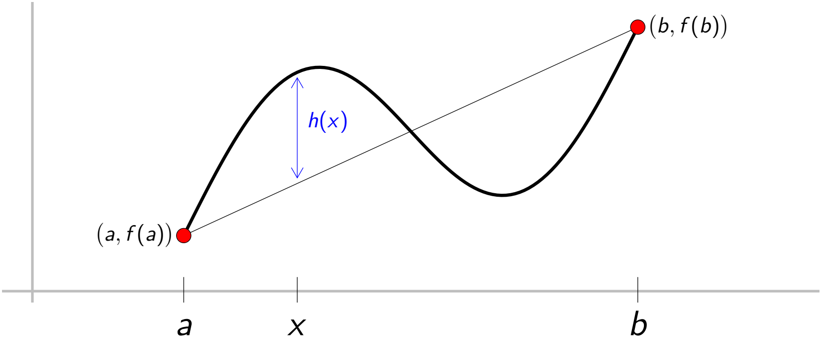

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Apply Rolle's theorem to

The Mean Value Theorem

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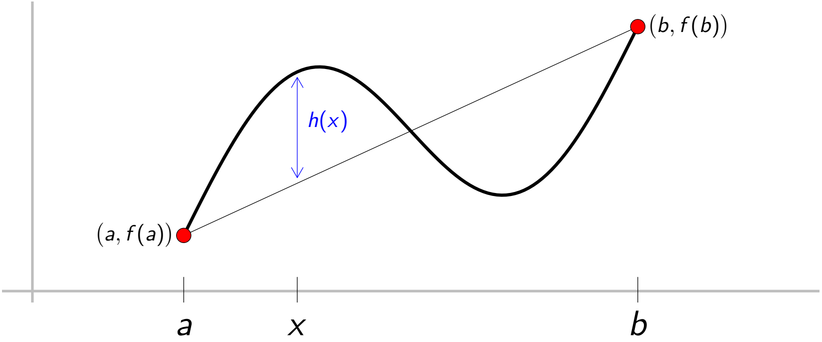

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Apply Rolle's theorem to

$$h(x)$$

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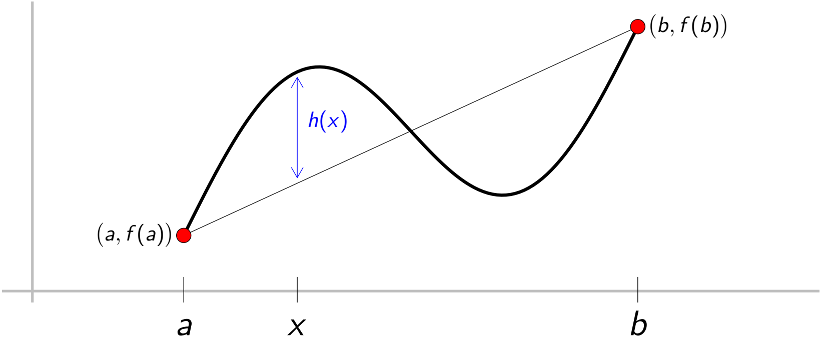

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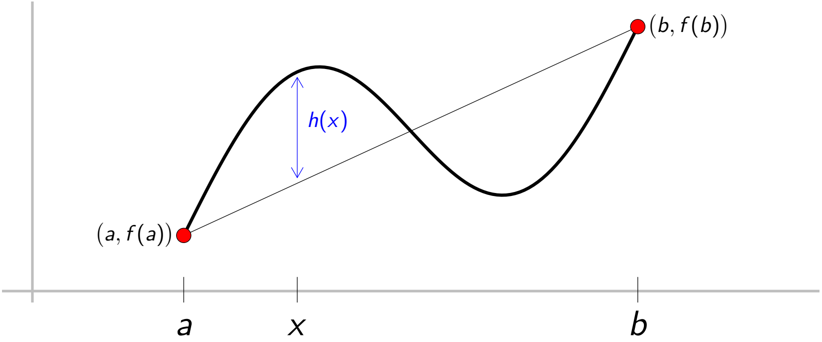
The Mean Value Theorem

Idea for proof:



Proof.

Apply Rolle's theorem to

$$h(x) = f(x)$$

The Mean Value Theorem

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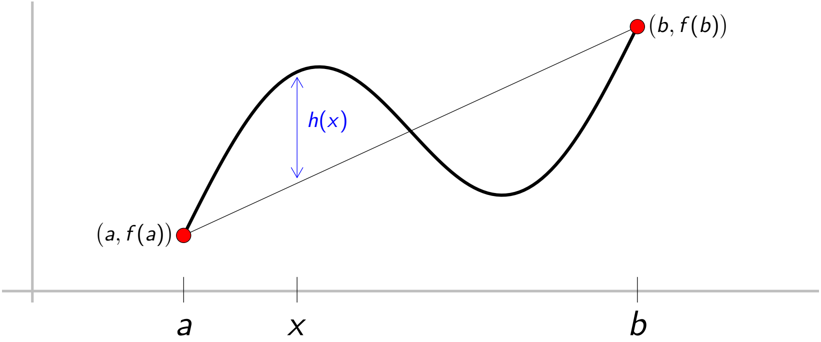

Proof.

Apply Rolle's theorem to

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The Mean Value Theorem

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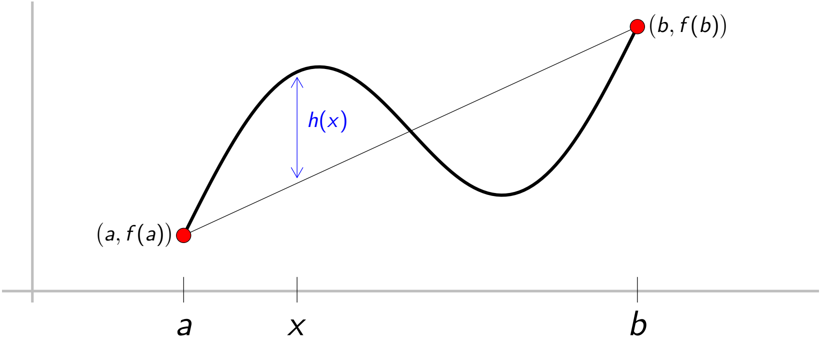


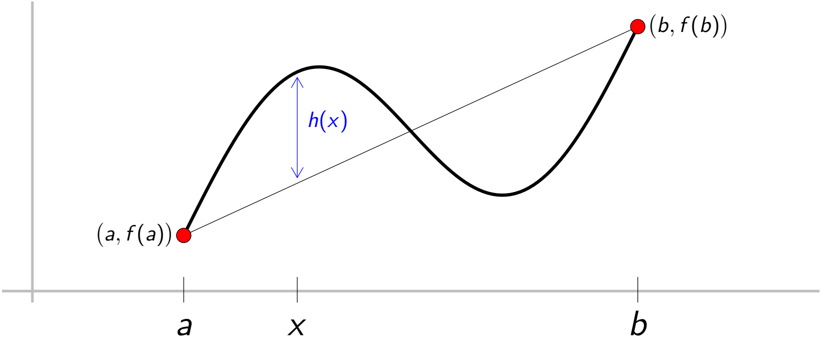
Proof.

Apply Rolle's theorem to

$$h(x) = f(x) - \Big[f(a)$$

The Mean Value Theorem

Idea for proof:



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The Mean Value Theorem

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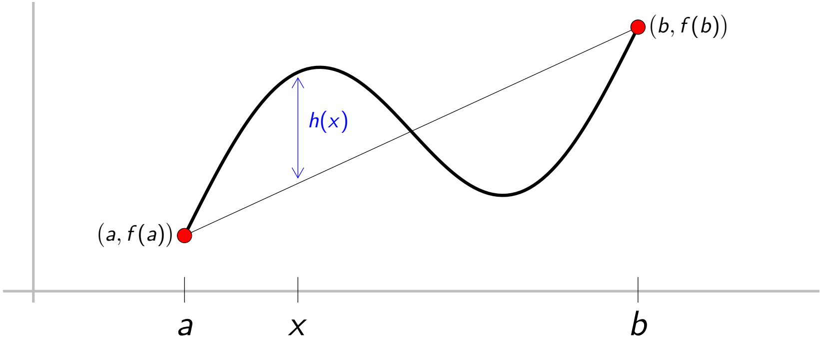

Proof.

Apply Rolle's theorem to

$$h(x) = f(x) - \left[f(a) + \left(\frac{f(b) - f(a)}{}\right.\right.$$

The Mean Value Theorem

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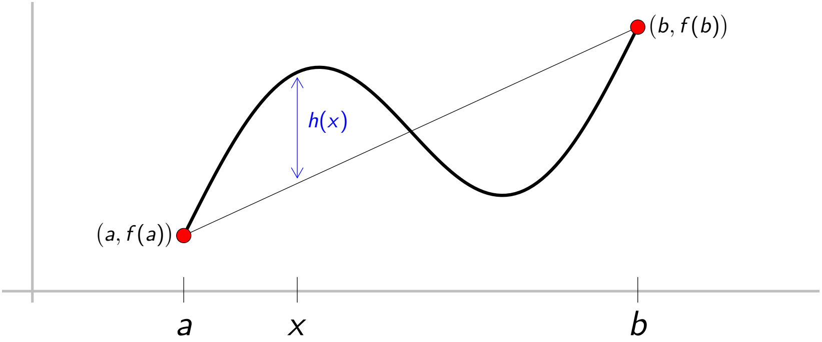

Proof.

Apply Rolle's theorem to

$$h(x) = f(x) - \left[f(a) + \left(\frac{f(b) - f(a)}{b - a}\right)\right.$$

The Mean Value Theorem

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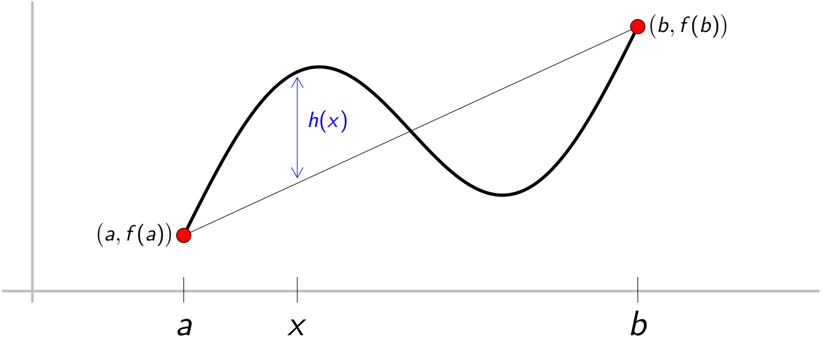

Proof.

Apply Rolle's theorem to

$$h(x) = f(x) - \left[f(a) + \left(\frac{f(b) - f(a)}{b - a}\right)(x - a)\right].$$

□

The Mean Value Theorem

The Mean Value Theorem

Example

$f'(x) > 0$ on an interval $I \implies f$ strictly increasing on I.

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Proof:

Suppose $x_1, x_2 \in I$

The Mean Value Theorem

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Proof:

Suppose $x_1, x_2 \in I$ and $x_1 < x_2$.

The Mean Value Theorem

Example

$f'(x) > 0$ on an interval $I \implies f$ strictly increasing on I.

Proof:

Suppose $x_1, x_2 \in I$ and $x_1 < x_2$. We must show

The Mean Value Theorem

Example

$f'(x) > 0$ on an interval $I \implies f$ strictly increasing on I.

Proof:

Suppose $x_1, x_2 \in I$ and $x_1 < x_2$. We must show $f(x_1) < f(x_2)$.

The Mean Value Theorem

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Suppose $x_1, x_2 \in I$ and $x_1 < x_2$. We must show $f(x_1) < f(x_2)$.

Since $f'(x)$ exists

The Mean Value Theorem

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Proof:

Suppose $x_1, x_2 \in I$ and $x_1 < x_2$. We must show $f(x_1) < f(x_2)$.

Since $f'(x)$ exists for all $x \in I$,

The Mean Value Theorem

Example

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Proof:

Suppose $x_1, x_2 \in I$ and $x_1 < x_2$. We must show $f(x_1) < f(x_2)$.

Since $f'(x)$ exists for all $x \in I$, f is certainly differentiable

The Mean Value Theorem

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Since $f'(x)$ exists for all $x \in I$, f is certainly differentiable on the closed subinterval

The Mean Value Theorem

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Since $f'(x)$ exists for all $x \in I$, f is certainly differentiable on the closed subinterval $[x_1, x_2]$.

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But $x_2 - x_1 > 0$

The Mean Value Theorem

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Since $f'(x)$ exists for all $x \in I$, f is certainly differentiable on the closed subinterval $[x_1, x_2]$.

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But $x_2 - x_1 > 0$ and since $x_* \in I$, we know $f'(x_*) > 0$.

$\therefore\ f(x_2) - f(x_1) > 0, \qquad$ *i.e.*, $\quad f(x_1) < f(x_2)$. $\square$

REMINDER: Intermediate Value Property

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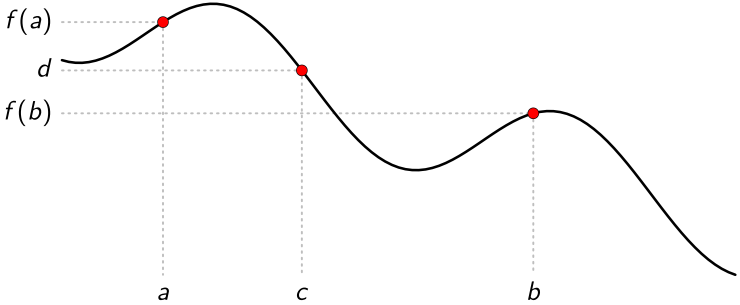

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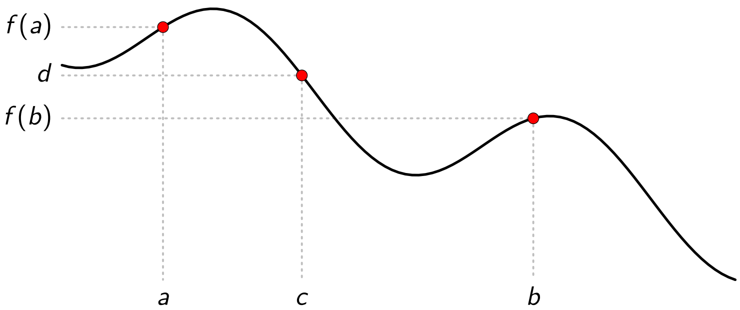
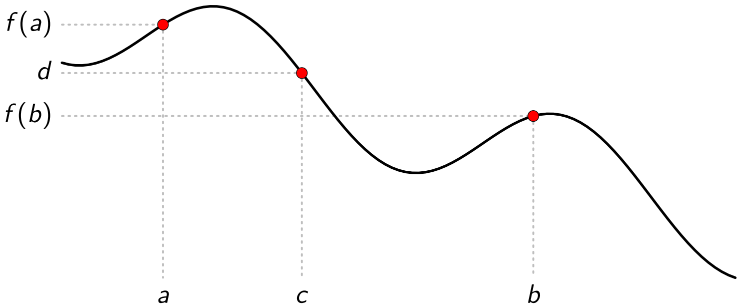


Definition (Intermediate Value Property (IVP))

REMINDER: Intermediate Value Property



Definition (Intermediate Value Property (IVP))

A function f

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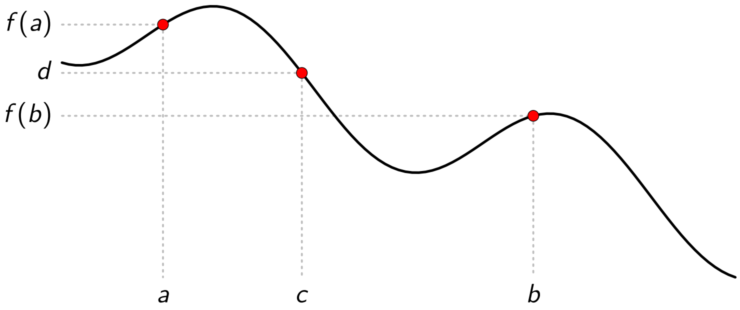

Definition (Intermediate Value Property (IVP))

A function f defined on

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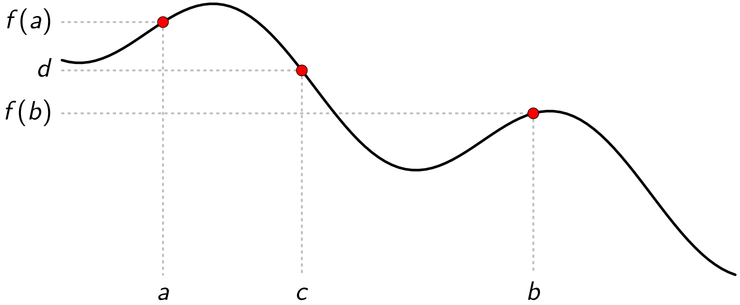
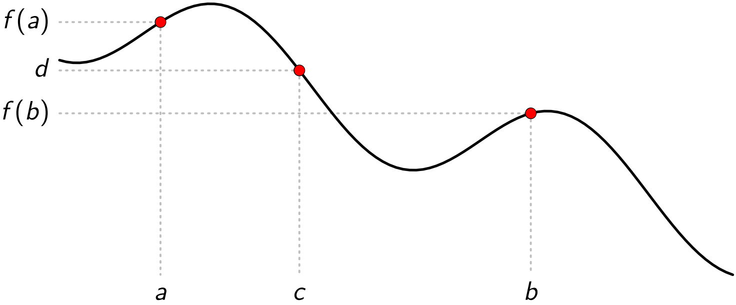


Definition (Intermediate Value Property (IVP))

A function f defined on an interval I

REMINDER: Intermediate Value Property



Definition (Intermediate Value Property (IVP))

A function f defined on an interval I is said to have

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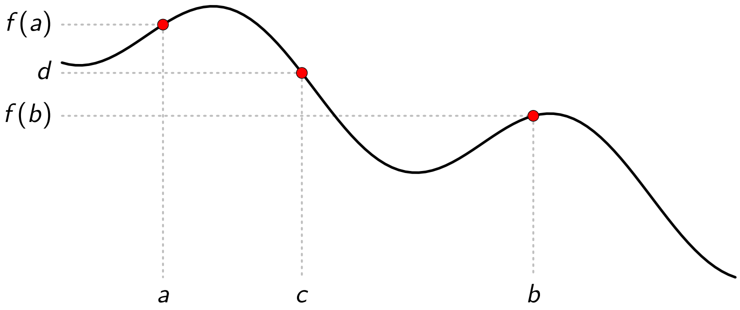
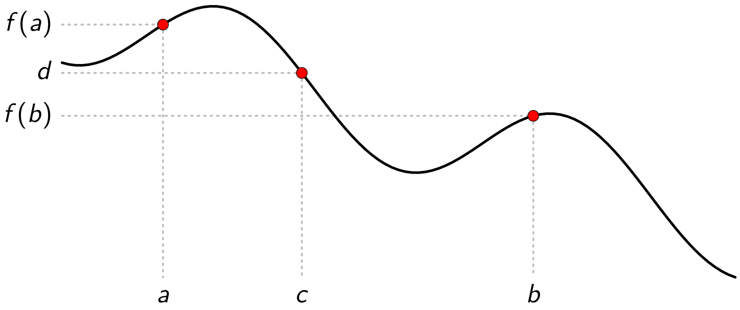


Definition (Intermediate Value Property (IVP))

A function f defined on an interval I is said to have the ***intermediate value property (IVP)***

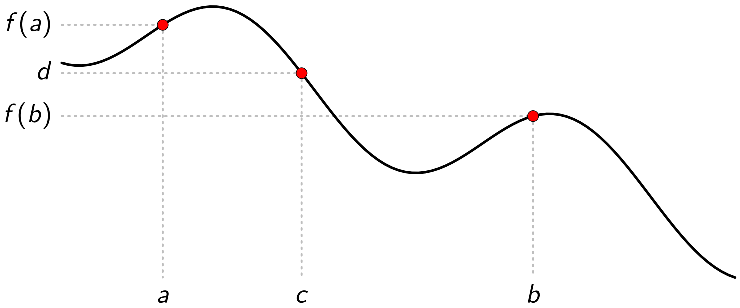
REMINDER: Intermediate Value Property



Definition (Intermediate Value Property (IVP))

A function f defined on an interval I is said to have the ***intermediate value property (IVP)*** on I

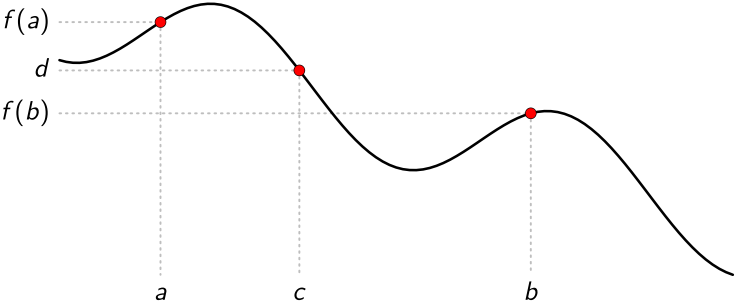
REMINDER: Intermediate Value Property



Definition (Intermediate Value Property (IVP))

A function f defined on an interval I is said to have the ***intermediate value property (IVP)*** on I iff

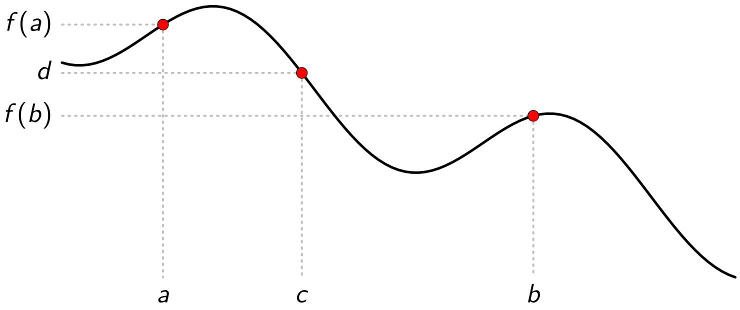
REMINDER: Intermediate Value Property



Definition (Intermediate Value Property (IVP))

A function f defined on an interval I is said to have the ***intermediate value property (IVP)*** on I iff for each $a, b \in I$

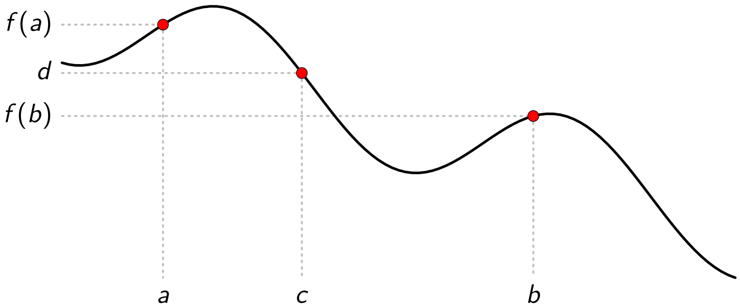
REMINDER: Intermediate Value Property



Definition (Intermediate Value Property (IVP))

A function f defined on an interval I is said to have the ***intermediate value property (IVP)*** on I iff for each $a, b \in I$ with $f(a) \neq f(b)$

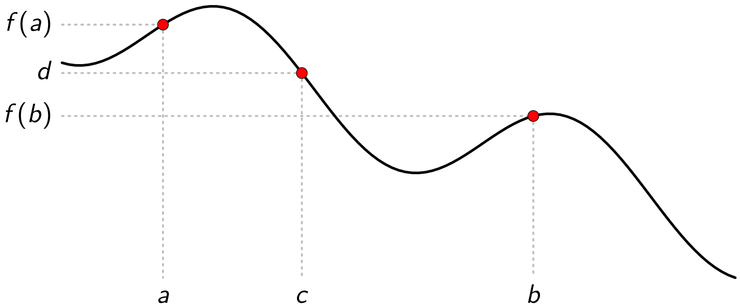
REMINDER: Intermediate Value Property



Definition (Intermediate Value Property (IVP))

A function f defined on an interval I is said to have the ***intermediate value property (IVP)*** on I iff for each $a, b \in I$ with $f(a) \neq f(b)$, and for each d

REMINDER: Intermediate Value Property



Definition (Intermediate Value Property (IVP))

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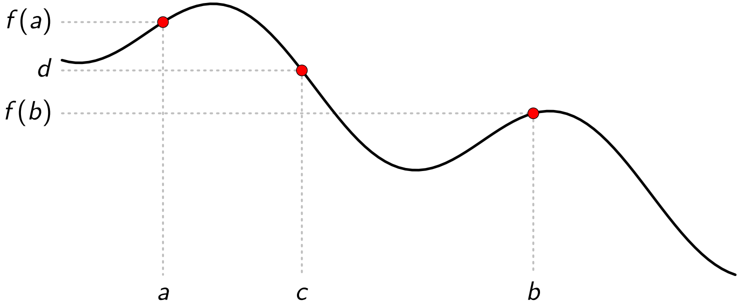
REMINDER: Intermediate Value Property



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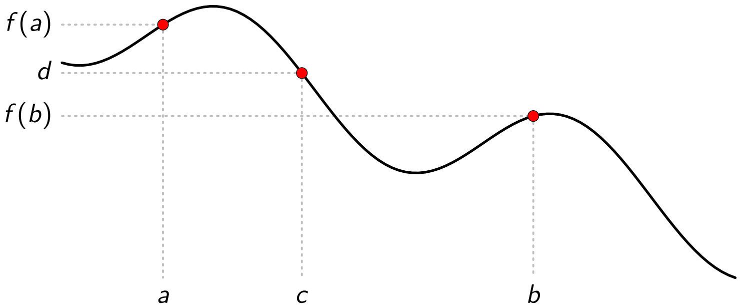
REMINDER: Intermediate Value Property



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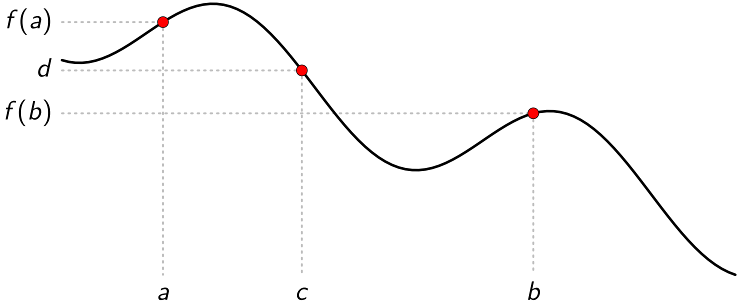
REMINDER: Intermediate Value Property



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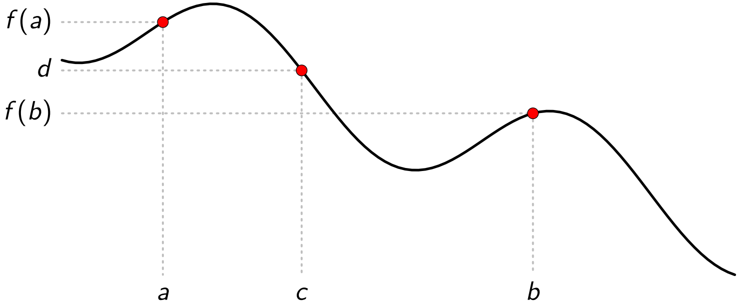
REMINDER: Intermediate Value Property



Definition (Intermediate Value Property (IVP))

A function f defined on an interval I is said to have the ***intermediate value property (IVP)*** on I iff for each $a, b \in I$ with $f(a) \neq f(b)$, and for each d between $f(a)$ and $f(b)$, there exists c between a and b for which $f(c) = d$.

REMINDER: Intermediate Value Property

REMINDER: Intermediate Value Property

Theorem (Intermediate Value Theorem (IVT))

REMINDER: Intermediate Value Property

Theorem (Intermediate Value Theorem (IVT))

If f is continuous

REMINDER: Intermediate Value Property

Theorem (Intermediate Value Theorem (IVT))

If f is continuous on an interval I

REMINDER: Intermediate Value Property

Theorem (Intermediate Value Theorem (IVT))

If f is continuous on an interval I then

REMINDER: Intermediate Value Property

Theorem (Intermediate Value Theorem (IVT))

If f is continuous on an interval I then f has

REMINDER: Intermediate Value Property

Theorem (Intermediate Value Theorem (IVT))

If f is continuous on an interval I then f has the intermediate value property (IVP)

REMINDER: Intermediate Value Property

Theorem (Intermediate Value Theorem (IVT))

If f is continuous on an interval I then f has the intermediate value property (IVP) on I.

REMINDER: Intermediate Value Property

Theorem (Intermediate Value Theorem (IVT))

If f is continuous on an interval I then f has the intermediate value property (IVP) on I.

Note:

REMINDER: Intermediate Value Property

Theorem (Intermediate Value Theorem (IVT))

If f is continuous on an interval I then f has the intermediate value property (IVP) on I.

Note: The interval I in the statement of the IVT

REMINDER: Intermediate Value Property

Theorem (Intermediate Value Theorem (IVT))

If f is continuous on an interval I then f has the intermediate value property (IVP) on I.

Note: The interval I in the statement of the IVT does not have to be closed

REMINDER: Intermediate Value Property

Theorem (Intermediate Value Theorem (IVT))

If f is continuous on an interval I then f has the intermediate value property (IVP) on I.

Note: The interval I in the statement of the IVT does not have to be closed and it does not have to be bounded.

REMINDER: Intermediate Value Property

Theorem (Intermediate Value Theorem (IVT))

If f is continuous on an interval I then f has the intermediate value property (IVP) on I.

Note: The interval I in the statement of the IVT does not have to be closed and it does not have to be bounded.

Unlike the extreme value theorem,

REMINDER: Intermediate Value Property

Theorem (Intermediate Value Theorem (IVT))

If f is continuous on an interval I then f has the intermediate value property (IVP) on I.

Note: The interval I in the statement of the IVT does not have to be closed and it does not have to be bounded.

Unlike the extreme value theorem, the IVT is not a theorem about functions defined on closed and bounded intervals.

Poll

- Go to
  https://www.childsmath.ca/childsa/forms/main_login.php
- Click on Math 3A03
- Click on Take Class Poll
- Fill in poll **Derivatives: IVP and derivatives**
- $\boxed{\texttt{Submit}}$.

Intermediate Value Property

Intermediate Value Property

Question:

Intermediate Value Property

Question: If a function has the IVP on an interval I, must it be continuous on I?

Intermediate Value Property

Question: If a function has the IVP on an interval I, must it be continuous on I?

Example

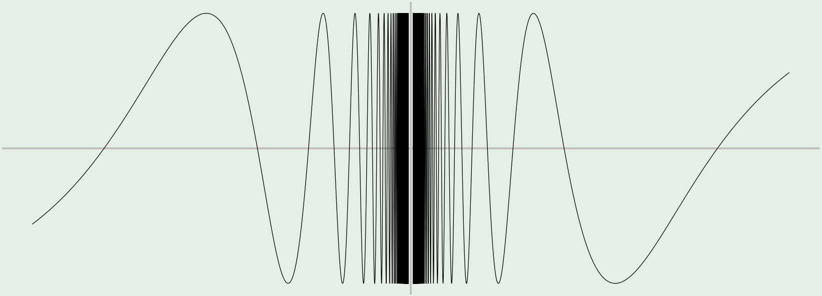
$$f(x) = \begin{cases} \sin \frac{1}{x} & x \neq 0, \\ 0 & x = 0. \end{cases}$$

Intermediate Value Property

Question: If a function has the IVP on an interval I, must it be continuous on I?

Example

$$f(x) = \begin{cases} \sin\frac{1}{x} & x \neq 0, \\ 0 & x = 0. \end{cases}$$



Intermediate value property for derivatives

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Theorem (Darboux's Theorem: IVP for derivatives)

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If f is differentiable

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Theorem (Darboux's Theorem: IVP for derivatives)

If f is differentiable on an interval I

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Theorem (Darboux's Theorem: IVP for derivatives)

If f is differentiable on an interval I then

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If f is differentiable on an interval I then its derivative

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Theorem (Darboux's Theorem: IVP for derivatives)

If f is differentiable on an interval I then its derivative f'

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- *Equivalent (contrapositive) statement of Darboux's theorem:* If a function does not have the intermediate value property on I then it is impossible that it is the derivative of any function

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Proof of Darboux's Theorem.

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Consider $a, b \in I$ with $a < b$.

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Consider $a, b \in I$ with $a < b$.
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Therefore, $x \in (a, b)$.

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Now suppose more generally that $f'(a) < K < f'(b)$.

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Now suppose more generally that $f'(a) < K < f'(b)$. Let $g(x) = f(x) - Kx$.

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Now suppose more generally that $f'(a) < K < f'(b)$. Let $g(x) = f(x) - Kx$. Then g is differentiable on I and $g'(x) = f'(x) - K$ for all $x \in I$.

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Now suppose more generally that $f'(a) < K < f'(b)$. Let $g(x) = f(x) - Kx$. Then g is differentiable on I and $g'(x) = f'(x) - K$ for all $x \in I$. In addition, $g'(a) = f'(a) - K < 0$

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Now suppose more generally that $f'(a) < K < f'(b)$. Let $g(x) = f(x) - Kx$. Then g is differentiable on I and $g'(x) = f'(x) - K$ for all $x \in I$. In addition, $g'(a) = f'(a) - K < 0$ and $g'(b) = f'(b) - K > 0$, so by the argument above, $\exists x \in (a, b)$ such that $g'(x) = 0$, *i.e.*, $f'(x) - K = 0$

Intermediate value property for derivatives

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The case $f'(a) > K > f'(b)$ is similar. $\square$

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Example ($f'(x) \neq 0 \; \forall x \in I \implies f \nearrow$ or $\searrow$ on I)

If f is differentiable on an interval I and $f'(x) \neq 0$ for all $x \in I$ then f is either increasing or decreasing on the entire interval I.

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The Derivative of an Inverse

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Example (Sufficient condition for *non*-differentiable inverse)

Suppose f is continuous and one-to-one on an interval I. If $x \in I$ and $f'(x) = 0$ then f^{-1} is <u>not</u> differentiable at $y = f(x)$.

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   i.e., $$(f^{-1})'(y) = \frac{1}{f'((f^{-1}(y))} \quad \text{for all } y \in J.$$

(TBB Theorem 7.32, p. 445)

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